

TUTORIAL 2 (FP2)

CHAPTER 2: HYPERBOLIC FUNCTIONS

Section I – Notes and Exercises

2.1: HYPERBOLIC FUNCTIONS AND THEIR GRAPHS**Hyperbolic Cosine and Hyperbolic Sine Functions**

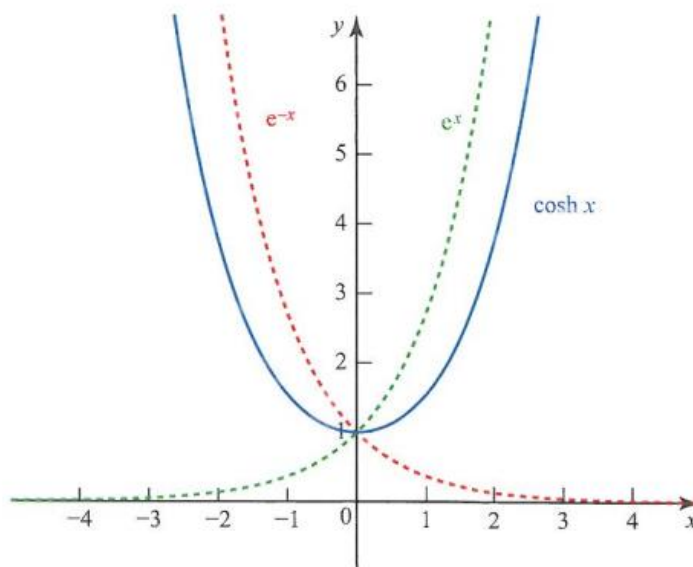
1. The hyperbolic cosine function, written $\cosh x$, is defined as below:

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

Avg of e^x and e^{-x}

Since $f(x) = f(-x)$, f is an even func.
 $\therefore$ Graph of $y = \cosh x$ is symmetrical about y -axis

The graph of $y = \cosh x$ is shown below.



Notice that:

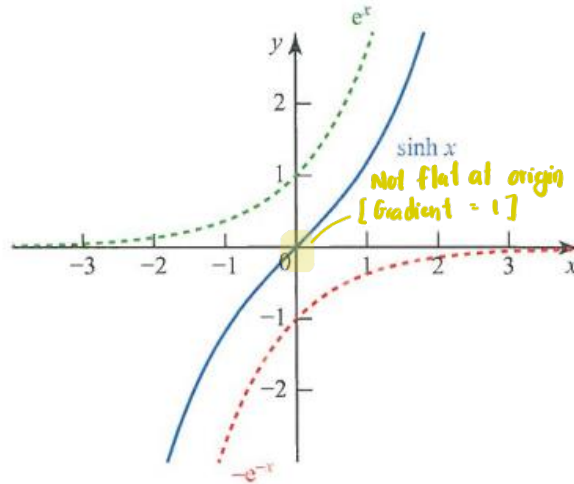
- The graph of $y = \cosh x$ lies **midway between the graphs of $y = e^x$ and $y = e^{-x}$** .
- The function has a **minimum point** at (0, 1).
- The (natural) domain of the cosh function is $x \in \mathbb{R}$ and its range is $y \geq 1$.
- The cosh function is an **even function**. (Do you remember what is meant by an “even function”?)
- The cosh function is **NOT a one-to-one function**.

2. The hyperbolic sine function, written $\sinh x$, is defined as below:

$$\sinh x = \frac{e^x - e^{-x}}{2} = \frac{e^x + (-e^{-x})}{2}$$

The graph of $y = \sinh x$ is shown below.

$$\begin{aligned} f(x) &= \frac{e^x - e^{-x}}{2} \\ f(-x) &= \frac{e^{-x} - e^{-(-x)}}{2} \\ &= \frac{e^{-x} - e^x}{2} \\ &= -\frac{(e^x - e^{-x})}{2} \\ &= -f(x) \end{aligned}$$



Notice that:

- The graph of $y = \sinh x$ lies **midway between the graphs of $y = e^x$ and $y = -e^{-x}$.**
- It **passes through the origin**, where it has a non-stationary **point of inflection**.
- The (natural) domain of the $\sinh$ function is $x \in \mathbb{R}$ and its range is $y \in \mathbb{R}$.
- The $\sinh$ function is an **odd function**. (Do you remember what is meant by an “odd function”?)
- The $\sinh$ function is a **one-to-one function**.

Other Hyperbolic Functions

1. The hyperbolic tangent function, written $\tanh x$, is defined as below:

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

- (i) By dividing top and bottom by e^x , we obtain

$$\tanh x = \frac{1 - e^{-2x}}{1 + e^{-2x}}$$

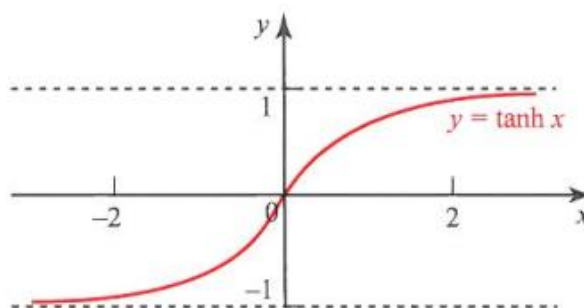
So, we see that as $x \rightarrow \infty, y \rightarrow 1$.

- (ii) By multiplying top and bottom by e^x , we obtain

$$\tanh x = \frac{e^{2x} - 1}{e^{2x} + 1}$$

So, we see that as $x \rightarrow -\infty, y \rightarrow -1$.

The graph of $y = \tanh x$ is shown below.



$$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

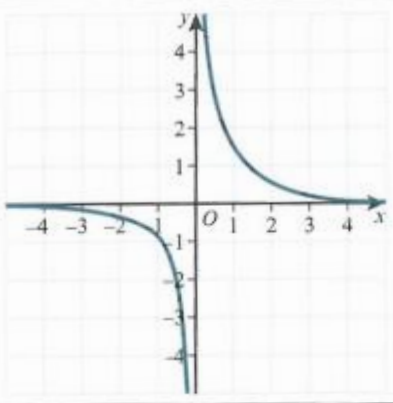
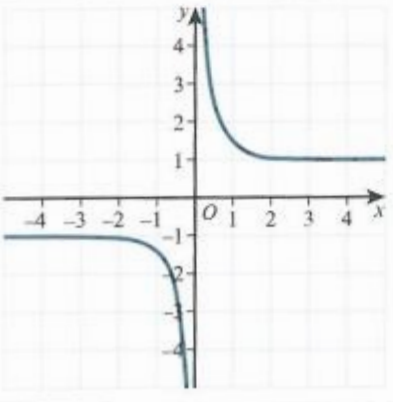
$$\begin{aligned} \tanh(-x) &= \frac{e^{-x} - e^x}{e^{-x} + e^x} \\ &= -\frac{(e^x - e^{-x})}{(e^x + e^{-x})} \\ &= -\tanh x \end{aligned}$$

Notice that:

- The (natural) domain of the $\tanh$ function is $x \in \mathbb{R}$ and the range is $-1 < y < 1$.
 - The $\tanh$ function is an **odd function**.
 - The $\tanh$ function is a **one-to-one function**.
2. The tables below show the graphs of the other three hyperbolic functions, namely

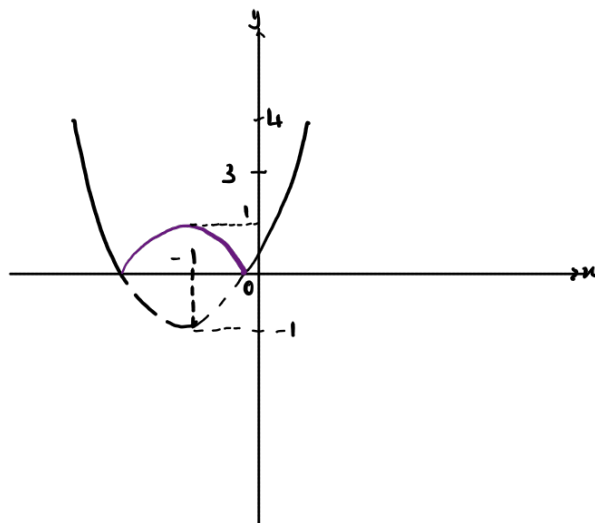
$$\operatorname{sech} x \equiv \frac{1}{\cosh x}, \quad \operatorname{cosech} x \equiv \frac{1}{\sinh x}, \quad \operatorname{coth} x \equiv \frac{1}{\tanh x}$$

Hyperbolic function	Pronunciation	Exponential form
$\operatorname{sech} x$	'setch'	$\operatorname{sech} x = \frac{2}{e^x + e^{-x}}$
		Domain: $x \in \mathbb{R}$
		Range: $0 < f(x) \leq 1$
		Even function

Hyperbolic function	Pronunciation	Exponential form
$\operatorname{cosech} x$	'cosetch'	$\operatorname{cosech} x = \frac{2}{e^x - e^{-x}}$
		Domain: $x \neq 0$
		Range: $f(x) \neq 0$
		Odd function
$\operatorname{coth} x$	'coth'	$\operatorname{coth} x = \frac{e^x + e^{-x}}{e^x - e^{-x}}$
		Domain: $x \neq 0$
		Range: $f(x) > 1, f(x) < -1$ OR
		Odd function

Example 1Sketch the graph of $y = |3 \cosh(x+1) - 4|$ **Solution**

$$\begin{array}{ll}
 \cosh x & f(x) \\
 \downarrow & \\
 \cosh(x+1) & f(x+1) \\
 \downarrow & \\
 3 \cosh(x+1) & 3f(x+1) \\
 \downarrow & \\
 3 \cosh(x+1) - 4 & 3f(x+1) - 4
 \end{array}$$



Example 2

Leaving your answer in exact form, solve $4 \cosh x - 3 \sinh x = 4$.

(Ans: $x = \ln 7$ or $x = 0$)

Solution

$$\begin{aligned}
 4 \cosh x - 3 \sinh x &= 4 \\
 4 \left(\frac{e^x + e^{-x}}{2} \right) - 3 \left(\frac{e^x - e^{-x}}{2} \right) &= 4 \\
 2e^x + 2e^{-x} - \frac{3}{2}e^x + \frac{3}{2}e^{-x} &= 4 \\
 \frac{1}{2}e^x + \frac{3}{2}e^{-x} - 4 &= 0 \\
 e^x + 3e^{-x} - 8 &= 0 \\
 (e^x)^2 - 8e^x + 3 &= 0
 \end{aligned}$$

$$\begin{aligned}
 (e^x - 7)(e^x - 1) &= 0 \\
 e^x &= 7 \text{ or } e^x = 1 \\
 x &= \ln 7 \text{ or } x = \ln 1 = 0
 \end{aligned}$$

Example 3

Solve $3 \tanh^2 x - 4 \tanh x = 4$

(Ans: $x = \tanh^{-1}\left(-\frac{2}{3}\right)$ only)

[Note: Later in this chapter, we will learn the **logarithmic forms** of inverse hyperbolic functions.]

Solution

$$\begin{aligned}
 3 \tanh^2 x - 4 \tanh x &= 4 \\
 3 \tanh^2 x - 4 \tanh x - 4 &= 0 \\
 (3 \tanh x + 2)(\tanh x - 2) &= 0 \\
 \tanh x &= -\frac{2}{3} \text{ or } \tanh x = 2 \\
 &\quad \text{(reject)} \\
 x &= \tanh^{-1}\left(-\frac{2}{3}\right) \quad (\because -1 < \tanh x < 1)
 \end{aligned}$$

2.2: HYPERBOLIC IDENTITIES

Proving Hyperbolic Identities

1. If we consider that hyperbolic functions are similar to trigonometric functions, it is reasonable to consider that hyperbolic identities exist in the same way. We can **use the exponential forms** to prove that they are true.

Example 4

Prove that $\cosh^2 x - \sinh^2 x \equiv 1$.

Solution

$$\begin{aligned}\cosh^2 x - \sinh^2 x &= \left(\frac{e^x + e^{-x}}{2} \right)^2 - \left(\frac{e^x - e^{-x}}{2} \right)^2 \\ &= \frac{(e^x + e^{-x})^2 - (e^x - e^{-x})^2}{4} \\ &= \frac{4(e^x)(e^{-x})}{4} \\ &= \frac{4(e^0)}{4} \\ &= 1\end{aligned}$$

$$\begin{aligned}(a+b)^2 - (a-b)^2 \\ &= a^2 + 2ab + b^2 - (a^2 - 2ab + b^2) \\ &= 4ab\end{aligned}$$

Example 5

Prove that $\sinh(A+B) \equiv \sinh A \cosh B + \cosh A \sinh B$.

Solution

$$\begin{aligned}\sinh A \cosh B + \cosh A \sinh B &= \left(\frac{e^A - e^{-A}}{2} \right) \left(\frac{e^B + e^{-B}}{2} \right) + \left(\frac{e^A + e^{-A}}{2} \right) \left(\frac{e^B - e^{-B}}{2} \right) \\ &= \frac{1}{4} \left[(e^A - e^{-A})(e^B + e^{-B}) + (e^A + e^{-A})(e^B - e^{-B}) \right] \\ &= \frac{1}{4} \left[(e^{A+B} + e^{A-B} - e^{-A+B} - e^{-A-B}) + (e^{A+B} - e^{A-B} + e^{-A+B} - e^{-A-B}) \right] \\ &= \frac{1}{4} \left[2e^{A+B} - 2e^{-A-B} \right] \\ &= \left[\frac{e^{A+B} - e^{-(A+B)}}{2} \right] \\ &= \sinh(A+B)\end{aligned}$$

Remark:

You should have noticed that the identities proven in Examples 4 and 5 are similar to trigonometric identities. In fact, trigonometric identities can be converted into hyperbolic identities using **Osborne's rule**. [see next page]

[**Warning:** The use of Osborne's rule does not constitute a proof! We can use Osborne's rule when we are required to use a hyperbolic identity without needing its proof.]

Osborne's Rule

- Osborne's rule states that, a trigonometric identity can be converted to an analogous identity for hyperbolic functions by
 - replacing trigonometric functions with their hyperbolic counterparts, AND THEN
 - flipping the sign** of each term involving the **product of two hyperbolic sines**.

- For example, given the identity

$$\cos(A - B) = \cos A \cos B + \sin A \sin B$$

Osborne's rule gives the corresponding identity

$$\cosh(A - B) = \cosh A \cosh B - \sinh A \sinh B$$

- Below are some examples of trigonometric identities with their corresponding hyperbolic identities.

Trigonometric identity	Hyperbolic identity
$\cos^2 \theta + \sin^2 \theta \equiv 1$	$\cosh^2 x - \sinh^2 x \equiv 1$
$\cos 2\theta \equiv \cos^2 \theta - \sin^2 \theta$	$\cosh 2x \equiv \cosh^2 x + \sinh^2 x$
$\sin 2\theta \equiv 2 \sin \theta \cos \theta$	$\sinh 2x \equiv 2 \sinh x \cosh x$

Example 6

Using Osborne's rule, state the angle addition formula for $\tanh(A + B)$.

[Hint: When applying Osborne's rule, $\tanh x$ should be interpreted as $\frac{\sinh x}{\cosh x}$.]

Solution

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

so need change sign

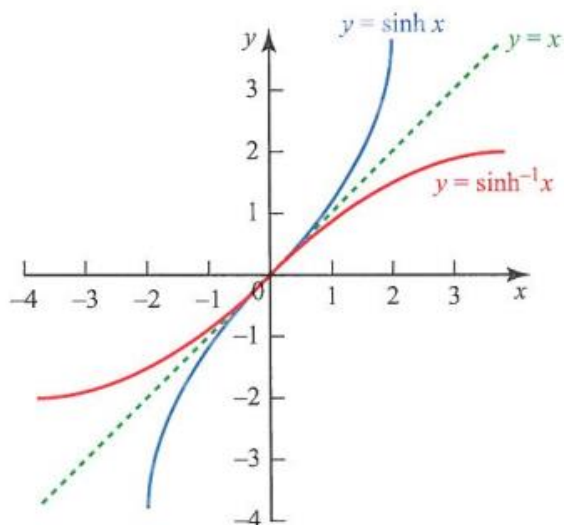
$$\tan A \tan B = \left(\frac{\sin A}{\cos A} \right) \left(\frac{\sin B}{\cos B} \right)$$

$$\therefore \tanh(A+B) = \frac{\tanh A + \tanh B}{1 + \tanh A \tanh B}$$

2.3: INVERSE HYPERBOLIC FUNCTIONS

Inverse of the Sinh and Tanh Functions

1. The inverse function of the sinh function is denoted by $\sinh^{-1}$.
2. The graph below shows the curves $y = \sinh x$ and $y = \sinh^{-1} x$.



3. As for any function and its inverse, the curves are reflections of each other in the line $y = x$.
4. We can find an expression (in **logarithmic form**) for $\sinh^{-1} x$, using the method below:

Let $y = \sinh^{-1} x$

Then $x = \sinh y$

$$\begin{aligned} x &= \frac{e^y - e^{-y}}{2} \\ 2x &= e^y - e^{-y} \\ e^y - e^{-y} - 2x &= 0 \\ e^{2y} - 1 - 2xe^y &= 0 \\ (e^y)^2 - 2x(e^y) - 1 &= 0 \end{aligned}$$

$$\begin{aligned} e^y &= \frac{2x \pm \sqrt{(-2x)^2 - 4(1)(-1)}}{2(1)} \\ &= \frac{2x \pm \sqrt{4x^2 + 4}}{2} \\ &= \frac{2x \pm 2\sqrt{x^2 + 1}}{2} \\ &= x \pm \sqrt{x^2 + 1} \\ &= x + \sqrt{x^2 + 1} \quad \text{or} \quad x - \sqrt{x^2 + 1} \end{aligned}$$

(reject)
($\therefore e^y > 0$)

When $x > 0$, $\begin{aligned} x^2 + 1 &> x^2 \\ \sqrt{x^2 + 1} &> \sqrt{x^2} \\ &> x \end{aligned}$ $\therefore x - \sqrt{x^2 + 1} < 0$	When $x < 0$ $\sqrt{x^2 + 1} > x$
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$$\begin{aligned} \therefore e^y &= x + \sqrt{x^2 + 1} \\ y &= \ln(x + \sqrt{x^2 + 1}) \\ \sinh^{-1} x &= \ln(x + \sqrt{x^2 + 1}) \quad (\text{proven}) \end{aligned}$$

So, $\sinh^{-1} x = \ln(x + \sqrt{x^2 + 1})$

5. You can find an expression (in **logarithmic form**) for the inverse of the tanh function in a similar way:

$$\text{Let } y = \tanh^{-1} x$$

$$\text{Then } x = \tanh y$$

$$x = \frac{e^{2y} - 1}{e^{2y} + 1}$$

$$xe^{2y} + x = e^{2y} - 1$$

$$(x-1)e^{2y} = -x-1$$

$$e^{2y} = \frac{-x-1}{1-x}$$

$$2y = \ln\left(\frac{1+x}{1-x}\right)$$

$$y = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right)$$

$$\boxed{\tanh^{-1} x = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right)} \quad (\text{proven})$$

$$\text{So, } \boxed{\tanh^{-1} x = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right)}$$

$$\begin{aligned} \tanh x &= \frac{\sinh x}{\cosh x} \\ &= \frac{e^x - e^{-x}}{e^x + e^{-x}} \\ &= \frac{e^{2x} - 1}{e^{2x} + 1} \end{aligned}$$

Additional Discussion

$$x = \tanh y$$

$$-1 < \tanh y < 1$$

$$\Rightarrow -1 < x < 1$$

$$\Rightarrow 0 < 1+x < 2 \quad \Rightarrow 1 > -x > -1$$

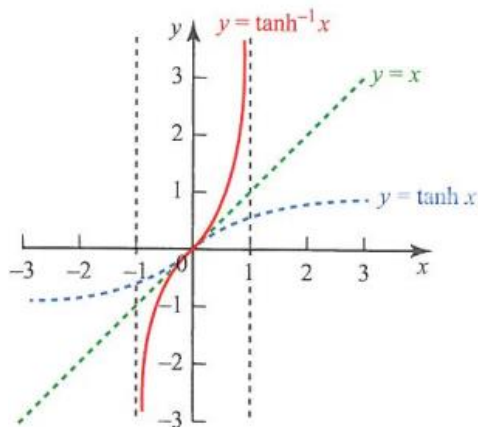
$$\therefore (1+x) \text{ is +ve} \quad 2 > 1-x > 0$$

$$0 < 1-x < 2$$

$$\therefore (1-x) \text{ is +ve}$$

Thus $\therefore \ln\left(\frac{1+x}{1-x}\right)$ is always defined when $-1 < x < 1$

6. The graph below shows the curves $y = \tanh x$ and $y = \tanh^{-1} x$.



7. Notice that both the **sinh function** and the **tanh function** are **one-to-one**, and so the inverse functions are defined over the whole domain.

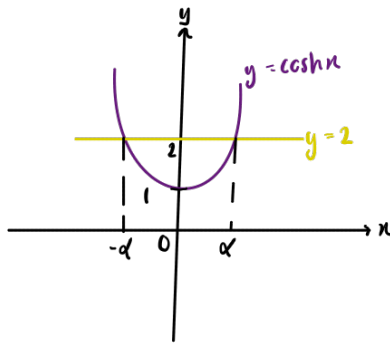
Inverse of the Cosh Function

1. Consider the following example:

Question:

Solve the equation $\cosh x = 2$.

Solution:



$$x = \ln(2 \pm \sqrt{3}) \quad x = \ln(2 + \sqrt{3}) \quad \text{or} \quad x = \ln(2 - \sqrt{3})$$

Method 1

$$\cosh x = 2$$

$$\frac{e^x + e^{-x}}{2} = 2$$

$$e^{2x} + 1 = 4e^x$$

$$(e^x)^2 - 4(e^x) + 1 = 0$$

$$e^x = \frac{4 \pm \sqrt{(-4)^2 - 4(1)(1)}}{2(1)}$$

$$= 2 \pm \sqrt{3}$$

Method 2

$$\cosh x = 2$$

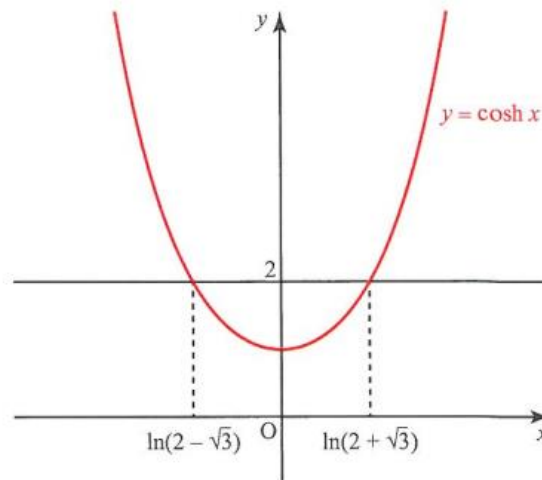
$$x = \pm \cosh^{-1}(2)$$

$$= \pm \ln(2 + \sqrt{4 - 1}) \quad [\text{from pg 11}]$$

$$= \pm \ln(2 + \sqrt{3})$$

$$= \ln(2 + \sqrt{3}) \quad \text{or} \quad \ln(2 - \sqrt{3})$$

2. The two roots of the equation in the example above are shown on the graph below.



You can see from the **symmetry** of the graph that **one of the roots is the negative of the other**.

Note:

It is probably not immediately obvious to you that $\ln(2 - \sqrt{3})$ is the negative of $\ln(2 + \sqrt{3})$. However, it is quite easy to prove that this is the case by adding them together.

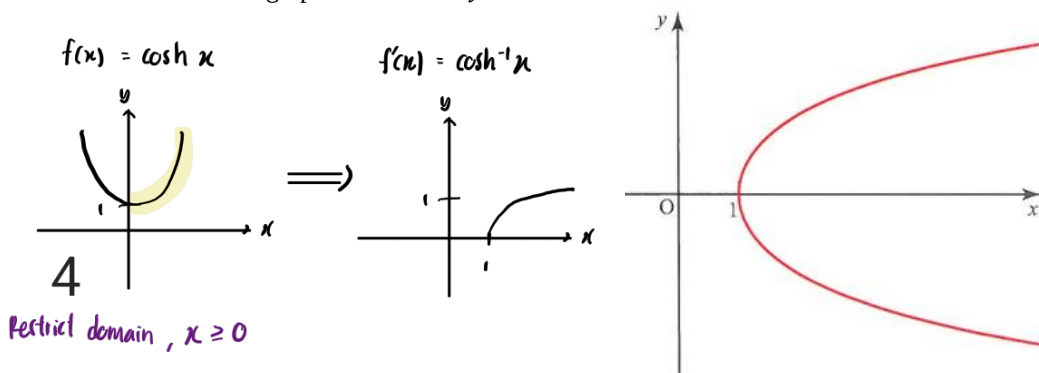
$$\ln(2 + \sqrt{3}) + \ln(2 - \sqrt{3}) = \ln((2 + \sqrt{3})(2 - \sqrt{3}))$$

$$= \ln[4 - 3]$$

$$= \ln 1$$

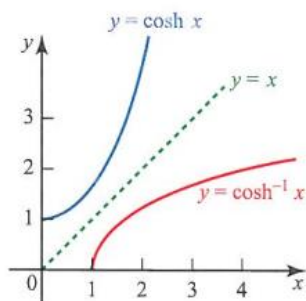
$$= 0$$

- The example on the previous page shows that the equation $\cosh x = k$ has two real roots if $k > 1$.
- The inverse of the cosh function is denoted by $\cosh^{-1}$. For this to be a function, $\cosh^{-1} k$ must have only one value.
- The graph of $x = \cosh y$ is shown below.



Notice that this graph goes above and below the x -axis, so the negative part has to be excluded to give the function $y = \cosh^{-1} x$.

- If the **domain** of the cosh function is **restricted to $x \geq 0$** , then the function is **one-to-one**. This restricted cosh function has an inverse function, the $\cosh^{-1}$ function.



- We will now derive the **logarithmic form** of $\cosh^{-1} x$.

$$\text{Let } y = \cosh^{-1} x$$

$$\text{Then } x = \cosh y$$

$$x = \frac{e^y + e^{-y}}{2}$$

$$e^{2y} - 2xe^y + 1 = 0$$

$$e^y = \frac{2x \pm \sqrt{(-2x)^2 - 4(1)(1)}}{2(1)}$$

$$e^y = x \pm \sqrt{x^2 - 1}$$

$$y = \ln(x + \sqrt{x^2 - 1}) \text{ or } \ln(x - \sqrt{x^2 - 1})$$

$$e^y = x + \sqrt{x^2 - 1} \quad \text{or} \quad e^y = x - \sqrt{x^2 - 1}$$

$$y = \ln(x + \sqrt{x^2 - 1}) \quad \text{or} \quad y = \ln(x - \sqrt{x^2 - 1})$$

(+ve value) (-ve value)

∴ Reject. $y \geq 0$

As in the example on the previous page, the second root is the negative of the first. Since $\cosh^{-1} x \geq 0$ by definition, the positive root is the required one.

Therefore: $\cosh^{-1} x = \ln(x + \sqrt{x^2 - 1})$.

Example 7

Solve the equation $7 \cosh^2 x + 2 \sinh^2 x = 14$, giving the answers in logarithmic form.

[Hint: $\cosh x$ cannot be negative, but $\sinh x$ can take positive and negative values.]

Solution

Method 1: Replace $\sinh^2 x$ by $(\cosh^2 x - 1)$

$$\cosh^2 x - \sinh^2 x = 1$$

$$(Ans: \pm \ln \left(\frac{4+\sqrt{7}}{3} \right))$$

$$7 \cosh^2 x + 2 \sinh^2 x = 14$$

$$7 \cosh^2 x + 2 (\cosh^2 x - 1) = 14$$

$$7 \cosh^2 x + 2 \cosh^2 x - 2 = 14$$

$$9 \cosh^2 x = 16$$

$$\cosh x = \pm \sqrt{\frac{16}{9}}$$

$$= \frac{4}{3} \text{ or } -\frac{4}{3}$$

$$(\because \cosh x \geq 1)$$

$$\cosh x = \frac{4}{3}$$

$$x = \pm \cosh^{-1} \left(\frac{4}{3} \right)$$

Method 2: Replace $\cosh^2 x$ by $(1 + \sinh^2 x)$

$$(Ans: \ln \left(\frac{4+\sqrt{7}}{3} \right))$$

$$7 \cosh^2 x + 2 \sinh^2 x = 14$$

$$7(1 + \sinh^2 x) + 2 \sinh^2 x = 14$$

$$9 \sinh^2 x = 7$$

$$\sinh x = \pm \frac{\sqrt{7}}{3}$$

$$x = \sinh^{-1} \left(\pm \frac{\sqrt{7}}{3} \right)$$

$$x = \ln \left(\frac{\sqrt{7}}{3} + \sqrt{\frac{7}{9} + 1} \right) \text{ or } x = \ln \left(-\frac{\sqrt{7}}{3} + \sqrt{\frac{7}{9} + 1} \right)$$

$$= \ln \left(\frac{\sqrt{7}}{3} + \frac{\sqrt{16}}{3} \right) = \ln \left(\frac{\sqrt{7} + 4}{3} \right)$$

$$= \ln \left(\frac{\sqrt{7} + 4}{3} \right)$$

Exercise

1. Solve the equation $4 \cosh^2 x - 19 \sinh x - 9 = 0$, giving your answers as exact logarithms.

$$(Ans: x = \ln \left(\frac{-1+\sqrt{17}}{4} \right) \text{ or } \ln(5 + \sqrt{26}))$$

2. Prove that $\sinh^{-1} A + \sinh^{-1} B \equiv \sinh^{-1}(A\sqrt{1+B^2} + B\sqrt{1+A^2})$.

***** END OF SECTION I *****

Exercise

1. Solve the equation $4 \cosh^2 x - 19 \sinh x - 9 = 0$, giving your answers as exact logarithms.

$$\left(\text{Ans: } x = \ln\left(\frac{-1+\sqrt{17}}{4}\right) \text{ or } \ln(5 + \sqrt{26}) \right)$$

$$4 \cosh^2 x - 19 \sinh x - 9 = 0$$

$$4(1 + \sinh^2 x) - 19 \sinh x - 9 = 0$$

$$4 \sinh^2 x - 19 \sinh x - 5 = 0$$

$$(4 \sinh x + 1)(\sinh x + 5) = 0$$

$$\sinh x = -\frac{1}{4} \quad \text{or} \quad \sinh x = 5$$

$$x = \ln\left(\frac{-1 + \sqrt{17}}{4}\right) \quad \text{or} \quad x = \ln(5 + \sqrt{26})$$

Differentiating Hyperbolic Functions

1. The standard results are:

(a) $\frac{d}{dx}(\cosh x) = \sinh x$	$\frac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \tanh x$
(b) $\frac{d}{dx}(\sinh x) = \cosh x$	$\frac{d}{dx}(\operatorname{cosech} x) = -\operatorname{cosech} x \coth x$
(c) $\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$	$\frac{d}{dx}(\coth x) = -\operatorname{cosech}^2 x$

Notice that these results are similar, but not quite the same, as the corresponding results for the trigonometric functions.

Recall	
$\frac{d}{dx}(\cos x) = -\sin x$	$\frac{d}{dx}(\sec x) = \sec x \tan x$
$\frac{d}{dx}(\sin x) = \cos x$	$\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$
$\frac{d}{dx}(\tan x) = \sec^2 x$	$\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$

2. Proof for the standard results:

$$\begin{aligned}
 \text{a) } \frac{d}{dx}(\cosh x) &= \frac{d}{dx} \left[\frac{1}{2}(e^x + e^{-x}) \right] = \frac{1}{2}(e^x - e^{-x}) = \sinh x \\
 \text{b) } \frac{d}{dx}(\sinh x) &= \frac{d}{dx} \left[\frac{1}{2}(e^x - e^{-x}) \right] = \frac{1}{2}(e^x + e^{-x}) = \cosh x \\
 \text{c) } \frac{d}{dx}(\tanh x) &= \frac{d}{dx} \left(\frac{\sinh x}{\cosh x} \right) = \frac{(\cosh x)(\cosh x) - (\sinh x)(\sinh x)}{\cosh^2 x} \\
 &= \frac{\cosh^2 x - \sinh^2 x}{\cosh^2 x} \\
 &= \frac{1}{\cosh^2 x} \\
 &= \operatorname{sech}^2 x \quad \text{H}
 \end{aligned}$$

D	I
$+ 5x$	$\cos(5x)$
$- 5$	$\frac{1}{5} \sin(5x)$
$- 0$	$-\frac{1}{25} \cos(5x)$
$x \sin(5x) + \frac{1}{5} \cos(5x)$	

$$\frac{\sinh(x)}{\cosh^2(x)} = \tanh(x) \operatorname{sech}(x)$$

Differentiating Inverse Hyperbolic Functions

1. The standard results are:

$$(a) \frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2-1}}$$

$$(b) \frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{x^2+1}}$$

$$(c) \frac{d}{dx}(\tanh^{-1} x) = \frac{1}{1-x^2}$$

You should recognise a similarity with the inverse trigonometric results on Pages 1 and 2.

2. **Proof for the standard results:**

[Note:

(a) It is possible to differentiate the logarithmic versions of $\cosh^{-1} x$ and $\sinh^{-1} x$, but it is easier to work with the hyperbolic functions themselves and their identities.

(b) The **gradients** of $y = \cosh^{-1} x$, $y = \sinh^{-1} x$ and $y = \tanh^{-1} x$ are **always positive**.]

a) let $y = \cosh^{-1} x$

$$\cosh y = x$$

$$\sinh y \left(\frac{dy}{dx} \right) = 1$$

$$\frac{dy}{dx} = \frac{1}{\sinh y}$$

$$= \frac{1}{\sqrt{\cosh^2 y - 1}}$$

$$\frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2 - 1}}$$

$$\cosh^2 y - \sinh^2 y = 1$$

$$\sinh y = \pm \sqrt{\cosh^2 y - 1}$$

Since grad. of $y = \cosh^{-1} x$ is always +ve, we take the +ve sqrt

b) $y = \sinh^{-1} x$

$$\sinh y = x$$

$$(\cosh y) \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\cosh y}$$

$$= \frac{1}{\sqrt{1 + \sinh^2 y}}$$

$$\frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{1 + x^2}}$$

$$\cosh^2 y - \sinh^2 y = 1$$

$$\cosh y = \pm \sqrt{1 + \sinh^2 y}$$

Since grad. of $y = \sinh^{-1} x$ is always +ve, we take the +ve sqrt

c) let $y = \tanh^{-1} x$

$$\tanh y = x$$

$$(\operatorname{sech}^2 y) \left(\frac{dy}{dx} \right) = 1$$

$$\frac{dy}{dx} = \frac{1}{\operatorname{sech}^2 y}$$

$$= \frac{1}{1 - \tanh^2 y}$$

$$\frac{d}{dx}(\tanh^{-1} x) = \frac{1}{1 - x^2}$$

